

popular brands of alkaline batteries. These batteries are low-cost, have a high energy density, and are readily available. But these discharge quickly and have high internal resistance.

Testing and measurement

While designing batteries, quality assurance and safety testing are necessary for approval from government organisations and safety regulatory agencies. All the cell parameters a design engineer would need may not always be possible to quantify by direct measurement. Qualification testing, for example, checking for electromagnetic compatibility (EMC) compliance, is done before it is approved for use in a robot.

To determine whether a battery can fulfil the power demand when put to use in expected conditions, load testing is done. For knowing the ambient conditions for batteries, their behaviour under exposure to varied heat and temperature levels, altitudes, humidity, and fire is studied. Mechanical tests include options like vibration test, impact test, crush tests, shock test, nail penetration tests, drop test, among other similar ones.

Protection from or tolerance to issues such as overcharge (time/voltage), over-discharge, short circuit, misuse, or abuse is essential to ensure the safety of users. When it comes to measuring battery parameters for the robots, Viswanathan says, "We run batteries on a test bench with a constant load, and we also run road tests with more power drain."

In a real environment, the parameters may be quite different from those calculated using assumptions. Testing is required to check the actual parameters during practical usage.

There are numerous devices available in the market for analysing, recording, and monitoring a battery's parameters. Instruments and software systems that can measure and manage the life cycle of individual cells keep the user informed and ensure that the robot does not stop working all of a sudden.

For instance, SBS is a battery manufacturer that offers different battery testing, monitoring, and maintenance solutions that satisfy industry standards, besides safety programs.

Calibration factors are built into any piece of test equipment due to differences in the aging profiles of the different battery chemistries.

Depending on the application, the device chosen for testing may just test the amount of charge present and/or its voltage output, or it may do a more comprehensive testing that keeps a tab on other parameters too. The most common way to measure a battery's current state, often seen in schools and universities also, is using a multimeter.

While we may gain information on the state of charge, we still cannot determine whether the required amount of current will be delivered when demanded. Performance monitoring needs to be done on the user's end to verify that a battery is performing as required when used in the application for which it was chosen.

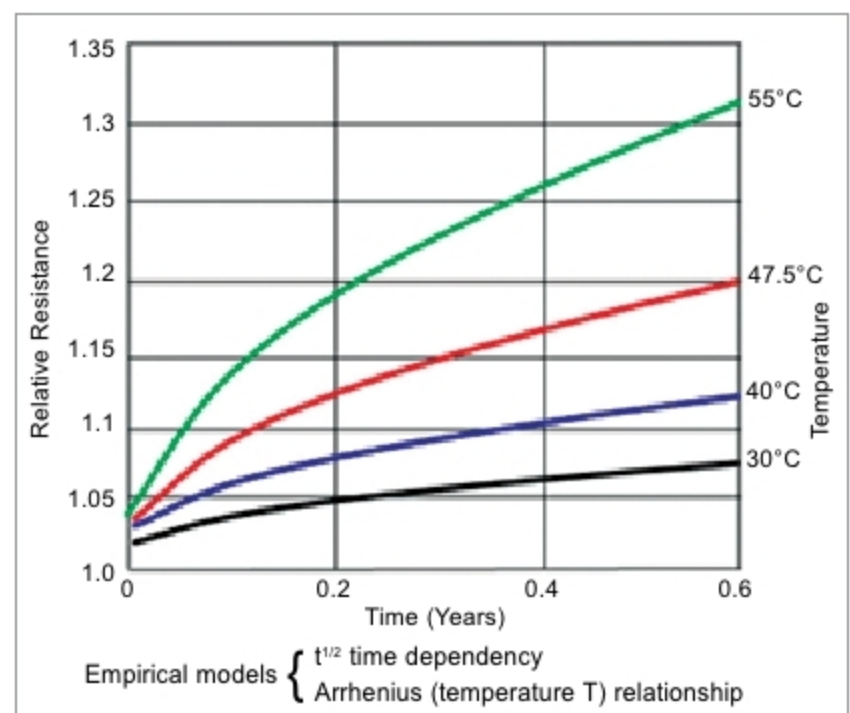
The result

There are several terms whose values give an insight into a battery's present condition once put to use. Whether or not to recharge or replace batteries depends on these values. Some of the parameters that different tools measure are explained next.

State of charge (SOC). It gives the present battery capacity as a percentage of maximum capacity. For many robotic applications, it is important to know the amount of energy left in a battery. It determines the change in battery capacity over time and is a crucial parameter in battery management systems.

State of health (SOH). It is a measure of a battery's ability to deliver the specified current when required. It is useful in understanding battery performance in real time during usage.

Depth of discharge (DOD). It is a measure of the battery capacity that has been discharged and is expressed as a percentage of maximum capacity. A discharge greater than or equal to eighty per cent is termed as a deep discharge.



Increasing resistance/impedance with time and temperature
(Credit: www.mpoweruk.com)

Terminal voltage. It is the voltage between the battery terminals when the actual load is applied. It depends on SOC and discharge/charge current.

Open-circuit voltage (OCV). It is the voltage between the battery terminals when no load is applied, that is, when the circuit is open. It determines the ability to deliver current but varies with SOC (directly proportional). After some time, as the battery begins to age, its ability to accept and to hold charge decreases. Hence, measuring internal resistance is necessary to get a complete picture of the situation.

Battery impedance (Z). A combination of internal resistance and reactance is known as impedance. The resistance within the battery is known as the internal resistance. It depends on SOC and is usually different for charging and discharging. The current generated by the battery interferes with direct measurement, hence ohmmeter is not used.

One method is to find the difference between terminal voltage and OCV, which gives IR voltage drop that corresponds to internal resistance. In another method, a small AC voltage of known frequency and amplitude is applied across the cells. Using Ohm's law, the value of impedance is calculated. The higher the internal impedance, the lower is the efficiency and the thermal stability, since more of the charging energy is converted into heat (I^2R). This indicates the SOH and measures deterioration in battery performance from the original. **EFY**